

Chapter 4 - System Design

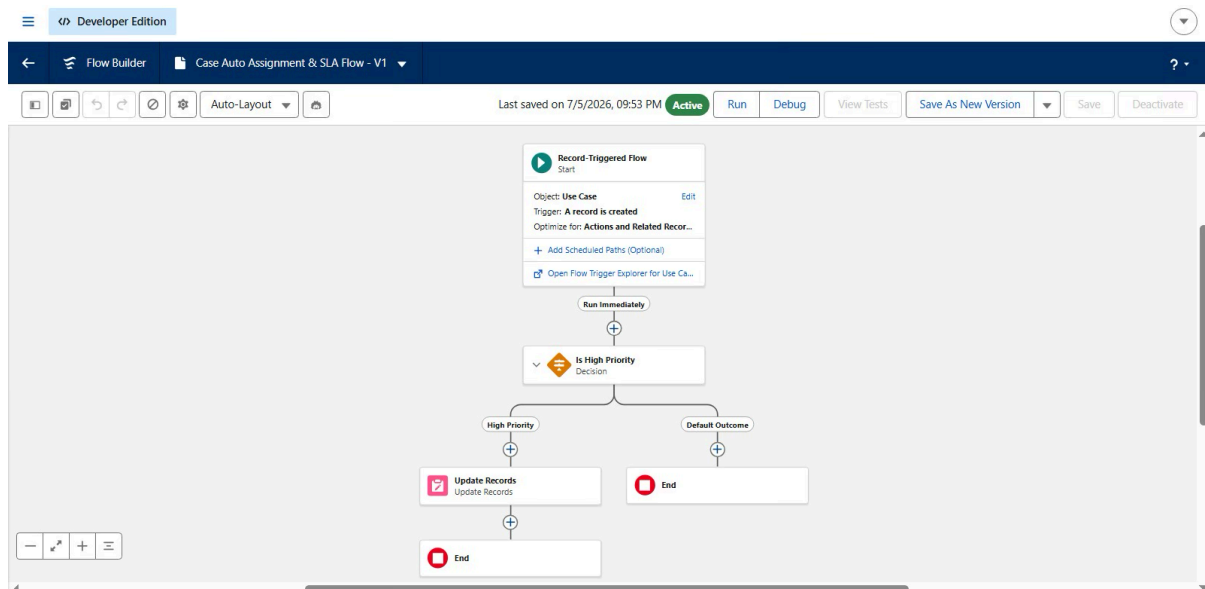
4.1 System Design Overview

The AI-Powered Customer Support & Service Automation system was designed using the Salesforce Lightning Platform. The design combines custom objects, lookup relationships, validation rules, record-triggered flows, queues, Prompt Builder, and Lightning App Builder to automate customer support processes while maintaining security and data integrity.

4.2 System Architecture

The application follows a layered architecture:

- **Presentation Layer:** Salesforce Lightning UI
- **Application Layer:** Flow Builder, Validation Rules, Prompt Builder
- **Data Layer:** Customer, Product, Order and Use Case objects
- **Security Layer:** Profiles, Queues and Field-Level Security
- **AI Layer:** Prompt Builder for AI-generated summaries



4.3 Custom Object Design

Object	Purpose
Customer	Stores customer information.
Product	Stores merchandise/product details.
Order	Stores customer purchase information.
Use Case	Stores customer support requests and AI summaries.

4.4 Customer Object Design

Field	Data Type
Customer Name	Text
Email	Email
Phone	Phone
Customer Type	Picklist

Developer Edition

Search Setup

SetupHomeObject Manager

Customer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

7 Items, Sorted by Field Label

Q Quick Find

New

Deleted Fields

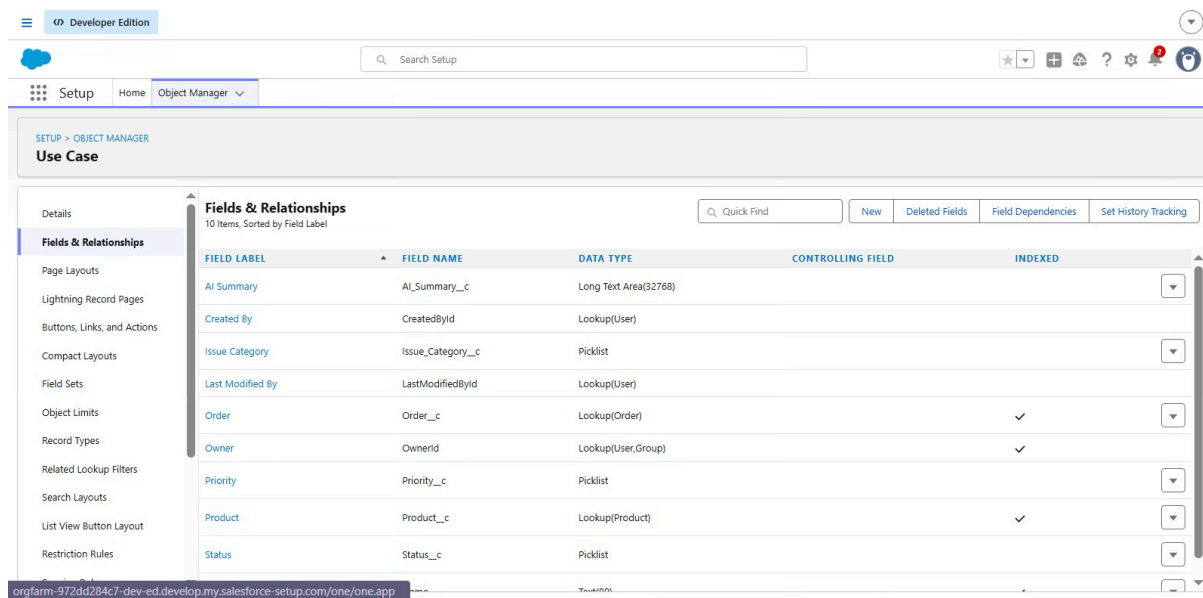
Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED	
Created By	CreatedById	Lookup(User)			
Customer Name	Name	Text(80)		✓	
Customer Type	Customer_Type__c	Picklist			
Email	Email__c	Email			
Last Modified By	LastModifiedById	Lookup(User)			
Owner	OwnerId	Lookup(User,Group)		✓	
Phone	Phone__c	Phone			

4.5 Product Object Design

Field	Data Type
Product Name	Text
Product Code	Text
Category	Picklist
Warranty (Months)	Number
Active	Checkbox



The screenshot shows the Salesforce Setup interface, specifically the 'Object Manager' section for a 'Use Case' object. The 'Fields & Relationships' tab is selected, displaying a list of 10 fields sorted by label. The fields are: AI Summary (Long Text Area), Created By (Lookup), Issue Category (Picklist), Last Modified By (Lookup), Order (Lookup), Owner (Lookup), Priority (Picklist), Product (Lookup), and Status (Picklist). The 'Indexed' column shows checkmarks for Order, Owner, Product, and Status. The interface includes a sidebar with navigation options like Details, Fields & Relationships, Page Layouts, and a top navigation bar with Setup, Home, and Object Manager.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
AI Summary	AI_Summary__c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Issue Category	Issue_Category__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Order	Order__c	Lookup(Order)		✓
Owner	OwnerId	Lookup(User,Group)		✓
Priority	Priority__c	Picklist		
Product	Product__c	Lookup(Product)		✓
Status	Status__c	Picklist		

4.6 Order Object Design

Field	Data Type
Order Number	Auto Number
Order Date	Date
Status	Picklist
Customer	Lookup (Customer)
Product	Lookup (Product)

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Order

Details

Fields & Relationships
9 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Customer	Customer__c	Lookup(Customer)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Order Date	Order_Date__c	Date		
Order Name	Name	Text(80)		✓
Order Number	Order_Number__c	Auto Number		
Owner	OwnerId	Lookup(User,Group)		✓
Product	Product__c	Lookup(Merchandise)		✓
Status	Status__c	Picklist		

orgfarm-972dd284c7-dev-ed.develop.my.salesforce-setup.com/one/one.app

4.7 Use Case Object Design

Field	Data Type
Product	Lookup (Product)
Order	Lookup (Order)
Priority	Picklist
Issue Category	Picklist
AI Summary	Long Text Area

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Use Case

Details

Fields & Relationships
10 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
AI Summary	AI_Summary__c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Issue Category	Issue_Category__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Order	Order__c	Lookup(Order)		✓
Owner	OwnerId	Lookup(User,Group)		✓
Priority	Priority__c	Picklist		
Product	Product__c	Lookup(Product)		✓
Status	Status__c	Picklist		

orgfarm-972dd284c7-dev-ed.develop.my.salesforce-setup.com/one/one.app

4.8 Object Relationship Design

Relationships implemented:

- Order → Customer (Lookup)
- Order → Product (Lookup)
- Use Case → Product (Lookup)
- Use Case → Order (Lookup)

These relationships ensure data consistency and eliminate redundancy.

The screenshot shows the Salesforce Setup interface in Developer Edition. The navigation menu on the left includes Setup, Home, and Object Manager. The main content area is titled 'Use Case' and displays the 'Use Case Validation Rule' configuration for the 'Use Case' object. The rule is named 'Damage_Requires_Priority' and is active. The error condition formula is: `AND( ISPICKVAL(Issue_Category__c,"Damage"), ISBLANK(TEXT(Priority__c)) )`. The error message is 'Priority is mandatory for Damage cases.' and the description is 'Ensures Priority is mandatory for Damage cases.' The rule was created by VELLAMPALLI VENKATANIHKIL on 7/6/2026 at 10:09 AM and was last modified by the same user at the same time.

Rule Name	Purpose	Object
Damage_Requires_Priority	Priority mandatory for Damage cases	Use Case
Warranty_Delivery_Requires_Priority	Priority mandatory for Warranty and Delivery cases	Use Case

4.9 Validation Rule Design

Rule Name	Purpose	Object
Damage_Requires_Priority	Priority mandatory for Damage cases	Use Case
Warranty_Delivery_Requires_Priority	Priority mandatory for Warranty and Delivery cases	Use Case

Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Use Case

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Use Case Validation Rule

Back to Use Case

Validation Rule Detail

Edit

Clone

Rule Name	Damage_Requires_Priority	Active	✓
Error Condition Formula	AND(ISPICKVAL(Issue_Category__c,"Damage"), ISBLANK(TEXT(Priority__c)))		
Error Message	Priority is mandatory for Damage cases.		
Description	Ensures Priority is mandatory for Damage cases.		
Created By	VELLAMPALLI VENKATANIKHIL AP24110010587, 7/6/2026, 10:09 AM	Modified By	VELLAMPALLI VENKATANIKHIL AP24110010587, 7/6/2026, 10:09 AM

Edit

Clone

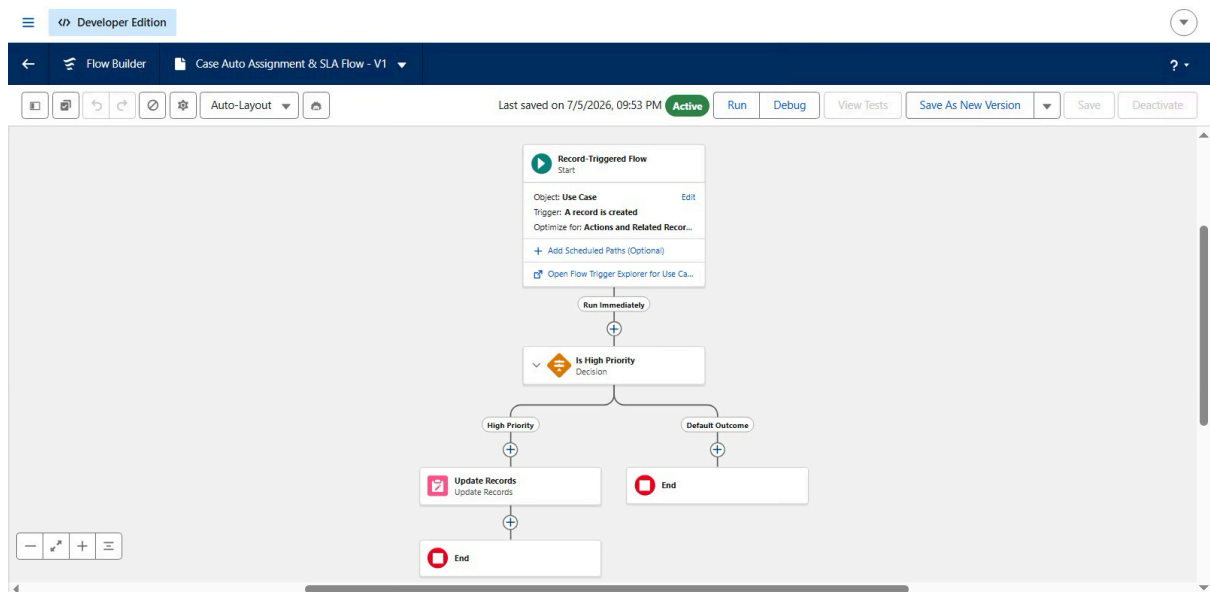
4.10 Flow Design

Flow 1: Case Assignment Flow

- Trigger: Record Created or Updated
- Object: Use Case
- Decision based on Issue Category.

Flow 2: Case Auto Assignment & SLA Flow

- Trigger: Record Created
- Checks Priority = High
- Assigns Owner to Damage Support Queue.



4.11 Queue Design

Queue	Purpose
Damage Support Queue	Handles high priority damage cases.
Warranty Support Queue	Handles warranty related support requests.

4.12 Prompt Builder Design

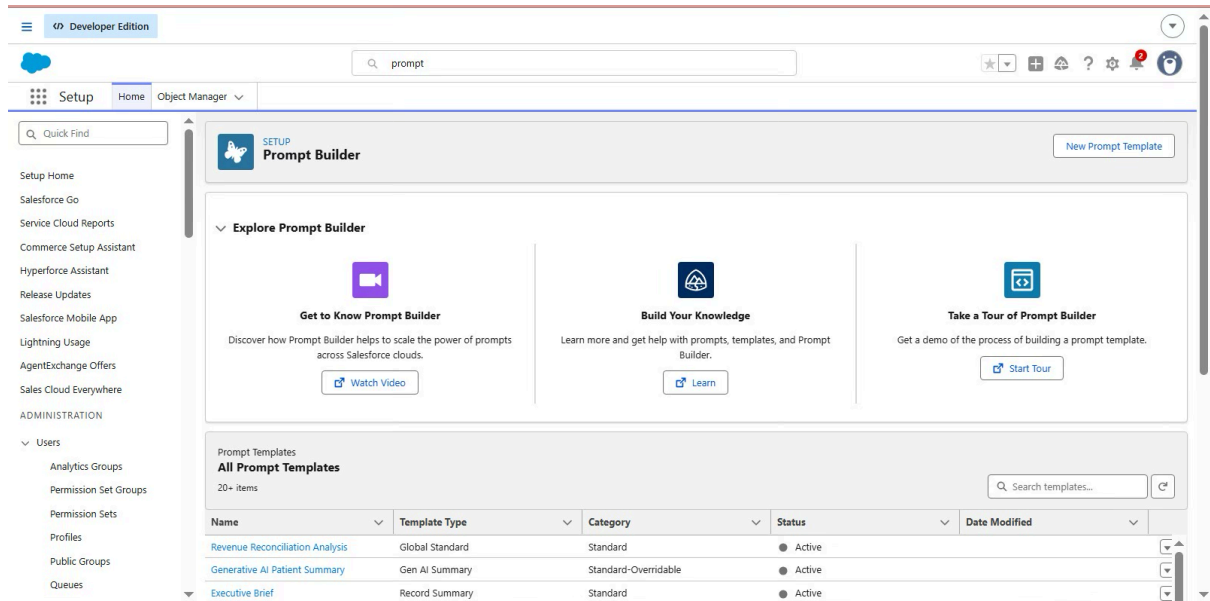
Prompt Template: MetadataInspection

Prompt Type: Field Generation

Object: Use Case

Output Field: AI Summary

Prompt Builder generates professional AI summaries for customer support cases using record information.

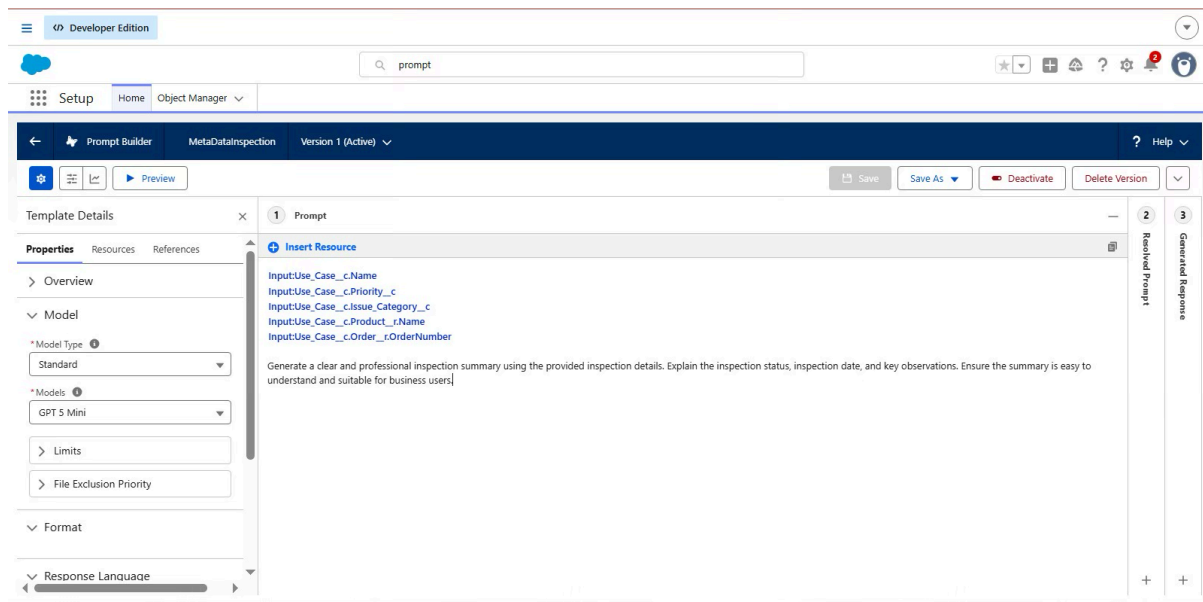


4.13 Lightning Application Design

Application Name: AI-Powered Customer Support & Service Automation

Navigation:

- Home
- Customer
- Product
- Order
- Use Case
- Reports
- Dashboards



4.14 Security Design

- Profiles
- Queues
- Field-Level Security
- Validation Rules
- Role-based access

4.15 Chapter Summary

The system design combines Salesforce data modeling, automation, AI integration, and security into a modular architecture. The design ensures scalability, maintainability, and efficient customer support management.